Additional Exercise 3 · Exam type

T3 Additional Exercise 3 – WIFINET S.L., statements 2017 and dividend options

Original statement · T3_Additional_EXERCISES.pdf, p. 4

Wifinet S.L. was incorporated on 1/01/2016 with a registered capital of 6,003 m.u., to provide wireless Internet on the Alicante coast.

Fiscal year 2016. It took a bank loan of 2,000 m.u. at 5% annual interest (payable at the end of each year), the principal being repayable in 2019. It invested 5,371 m.u. in technical equipment with a life of 20 years; at the end of 2016 it recorded an accumulated depreciation of 26 m.u. for those assets. It also invested 206 m.u. in licence fees by 31/12/2016, to use certain technologies for 5 years. It spent 636 m.u. on financial shares of other companies. Most of 2016 was devoted to building the network and business started only in the last two months. At the end of 2016 the balance sheet reflected a **loss of 340 m.u.** and **cash in hand and bank of 1,476 m.u.**

In 2017 the results were positive, but the company decided not to distribute dividends and to retain all its profits to reduce the loss of the previous year.

Fiscal year 2017.

- Internet and VOIP: 5,000 customers on average, with an average revenue of 0.032 m.u./month per client in subscription fees. Additionally, 46 m.u. of on-line sales of other products. At the end of 2017 customers owed 164 m.u.
- Stocks were exhausted at the end of 2016. Purchases of materials to be installed in customers' homes: 680 m.u., of which 560 m.u. paid immediately and the rest deferred to the following year. The 2017 sales consumed half of the purchased goods.
- Staff costs: 559 m.u., paid in full in 2017. Supplies and outside services: 312 m.u., paid.
- Other administrative expenses: 300 m.u., 60% of them deferred to the following year.
- Corporation tax: 30%, payable in the first quarter of 2018.

Required: (1) the income statement for 2017; (2) the balance sheet at 31/12/2017; (3) which balance-sheet items change, and by how much, if a dividend of 20% of net profit is paid at 24:00 on 31/12/2017; (4) the same if the dividend of 20% is decided before the end of 2017 and paid on 30/06/2018.

Phase 1 — Setup: data, timeline and hypotheses

Period: calendar year 2017. **Closing:** 31/12/2017. The 2016 data serve only to build the **opening balance** of 2017.

Hypotheses:

1. The licences were bought at the very end of 2016, so they start to be amortised in 2017 (no 2016 amortisation is given).
2. The equipment was depreciated only 26 m.u. in 2016 (part of the year); in 2017 it is used the whole year.
3. Interest is paid at the end of each year, so no interest payable remains; the loan is due in 2019, more than one year after 31/12/2017: **long-term** [Slide 16].
4. The financial shares are a long-term investment (non-current asset) [Slide 15]; they produce no revenue in the data.

Phase 2 — Frames: opening balance and classification of every operation

Opening balance at 31/12/2016 (rebuilt from the data)

Assets	**m.u.**	**Liabilities and equity**	**m.u.**
Technical equipment ($5{,}371 - 26$)	5,345	Capital	6,003
Licences	206	Loss of 2016	(340)
Financial shares	636	**Equity**	**5,663**
Cash	1,476	Bank loan (due 2019)	2,000
Total	**7,663**	**Total**	**7,663**

It balances ($7{,}663 = 5{,}663 + 2{,}000$), which confirms that no other 2016 item is missing.

Operations of 2017

Operation	Income statement 2017	Cash 2017	Open at 31/12/2017
Subscriptions + on-line sales	Revenue 1,966	+1,802	Receivables 164
Purchases of materials (680)	COGS 340	−560	Inventory 340; suppliers 120
Staff	559	−559	–
Supplies and outside services	312	−312	–
Administrative expenses	300	−120	Other creditors 180
Equipment depreciation	268.55	0	Accumulated depreciation 294.55
Licence amortisation	41.20	0	Licences 164.80
Interest	100	−100	–
Tax	13.575	0	Taxes payable 13.575

Phase 3 – Derivation

Step 1. Revenues and receivables

$$\text{Subscriptions} = 5{,}000 \text{ clients} \times 0.032 \tfrac{\text{m.u.}}{\text{client}\cdot\text{month}} \times 12 \text{ months} = 1{,}920 \text{ m.u.}$$

$$\text{Revenues} = 1{,}920 + 46 = 1{,}966 \text{ m.u.}, \qquad \text{Collected} = 1{,}966 - 164 = 1{,}802 \text{ m.u.}$$

Step 2. Purchases, consumption and inventory

Opening stock 0 ("stocks were exhausted at the end of 2016"); purchases 680; half consumed:

$$\text{COGS} = 0 + 680 - 340 = 340 \text{ m.u.}, \qquad \text{Ending inventory} = 340 \text{ m.u.}$$

Payment: 560 paid; $680 - 560 = 120$ owed to suppliers.

Step 3. Operating expenses

- Staff 559 and supplies and outside services 312, all paid.
- Administrative expenses 300: 60% deferred, $0.60 \times 300 = 180$ owed; $300 - 180 = 120$ paid.

Step 4. Depreciation and amortisation

$$D_{\text{equipment}} = \frac{5{,}371}{20} = 268.55 \text{ m.u.}, \qquad A_{\text{licences}} = \frac{206}{5} = 41.20 \text{ m.u.}$$

Accumulated depreciation of the equipment: $26 + 268.55 = 294.55$; net value $5{,}371 - 294.55 = 5{,}076.45$. Licences: $206 - 41.20 = 164.80$.

Step 5. Interest

$$I = 2{,}000 \times 0.05 = 100 \text{ m.u.} \quad \text{(paid at the end of 2017)}$$

Step 6. Income statement 2017 (requirement 1)

Line	Computation	m.u.
Revenues	$1{,}920 + 46$	1,966.000
– Cost of goods sold (materials consumed)		340.000
= Gross profit		**1,626.000**
– Staff		559.000
– Supplies and outside services		312.000
– Administrative expenses		300.000
– Depreciation and amortisation	$268.55 + 41.20$	309.750
= Income from operations	$1{,}626 - 1{,}480.75$	**145.250**
– Interest expense		100.000
= Income before taxes	$145.25 - 100$	**45.250**
– Income taxes (30%)	0.30×45.25	13.575
= Net income	$45.25 - 13.575$	**31.675**

Step 7. Cash at 31/12/2017

$$\text{Cash} = 1{,}476 + 1{,}802 - 560 - 559 - 312 - 120 - 100 = 1{,}627 \text{ m.u.}$$

Step 8. Balance sheet at 31/12/2017 (requirement 2)

All the profit is retained, so it stays in equity and reduces the accumulated loss: $-340 + 31.675 = -308.325$.

Assets	m.u.	Liabilities and equity	m.u.
Technical equipment	5,371.000	Capital	6,003.000
– Accumulated depreciation	(294.550)	Loss of 2016	(340.000)
Licences (net)	164.800	Net income 2017	31.675
Financial shares	636.000	**Equity**	**5,694.675**
Non-current assets	**5,877.250**	Bank loan (due 2019)	2,000.000
Inventory	340.000	**Long-term liabilities**	**2,000.000**
Accounts receivable	164.000	Suppliers	120.000
Cash	1,627.000	Other creditors (administration)	180.000
Current assets	**2,131.000**	Taxes payable (Q1 2018)	13.575
		Current liabilities	**313.575**
Total assets	**8,008.250**	**Total liabilities and equity**	**8,008.250**

Step 9. Dividend paid at 24:00 on 31/12/2017 (requirement 3)

$$\text{Dividend} = 0.20 \times 31.675 = 6.335 \text{ m.u.}$$

Paying it at 24:00 means cash leaves the company and the profit kept in equity falls by the same amount:

Item	Before	Change	After
Cash	1,627.000	–6.335	1,620.665
Equity (net income retained)	5,694.675	–6.335	5,688.340
Total assets = total liabilities and equity	8,008.250	–6.335	8,001.915

Step 10. Dividend decided in 2017, paid on 30/06/2018 (requirement 4)

The dividend is already owed to the shareholders at 31/12/2017 but no cash has left: equity falls and a **current liability** (dividends payable, due within the year) appears.

Item	Before	Change	After
Equity	5,694.675	–6.335	5,688.340
Dividends payable (current liability)	0	+6.335	6.335
Current liabilities	313.575	+6.335	319.910
Cash and total assets	unchanged	0	8,008.250

Phase 4 — Interpretation and checks

- **Balance:** $5{,}877.25 + 2{,}131 = 8{,}008.25 = 5{,}694.675 + 2{,}000 + 313.575$; requirement 3: $8{,}001.915 = 5{,}688.340 + 2{,}313.575$; requirement 4: $8{,}008.25 = 5{,}688.34 + 2{,}000 + 319.91$.
- **Requirement 3 vs 4:** both reduce equity by the same 6.335; the difference is only on the other side: in 3 an **asset** (cash) falls, in 4 a **liability** rises. A dividend is never an expense, so net income is 31.675 in all three cases.
- **Why the company chose not to pay:** after 2017 the equity is still below the capital (accumulated loss -308.325); retaining the profit is self-financing [Slide 23] that rebuilds the equity.
- **Exam traps:** (1) the 2016 data are only the opening balance; (2) inventory is half of the purchases; (3) the licences bought at the end of 2016 are amortised from 2017; (4) the loan due in 2019 is long-term.
- No official numerical solution is included in `sources/`. Three decimals are kept because 268.55, 41.20 and 13.575 are exact.

Study notes written from the official course material (T3_Additional_EXERCISES.pdf; Topic 3 Financial Management (I).pdf · p. 7). Online version: subjects/business-management/problemas/topic-3-extra-exam-problems.html